



D427 Practice Test 1



1. Which of the following functions would be used to combine two columns:
 - a. TRUNCATE()
 - b. AGGREGATE()
 - c. SUBSTRING()
 - d. CONCAT()

2. The Book table has the following columns:
genre – varchar(20)
pages – integer
author_id – char(3)
isbn_number – varchar(20)
Which column should be designated at the **primary key** for the Book table?
 - a. genre
 - b. pages
 - c. author_id
 - d. isbn_number

3. The Book table has the following columns:
genre – varchar(20)
pages – integer
author_id – char(3)
isbn_number – varchar(20)
Which column should be designated as the **foreign key** for the Book table?
 - a. genre
 - b. pages
 - c. author_id
 - d. isbn_number

4. Which data type represents numbers with fractional values:
 - a. Integer
 - b. Decimal
 - c. Character
 - d. Binary

5. Which of the following is a DDL commands?

- a. INSERT
- b. SELECT
- c. CREATE INDEX
- d. UPDATE

6. Which of the following is a DML command?

- a. CREATE VIEW
- b. CREATE TABLE
- c. INSERT
- d. ALTER INDEX

7.

```
CREATE TABLE Invoice (  
  invoice_id      INT      NOT NULL  AUTO_INCREMENT,  
  date            DATE     NOT NULL,  
  customer_id     INT      NOT NULL  UNIQUE,  
  PRIMARY KEY (invoice_id),  
  FOREIGN KEY (customer_id) REFERENCES Customer (customer_id) ON DELETE  
  CASCADE  
);
```

Looking at the Customer and Invoice tables and the CREATE TABLE for the Invoice table with foreign key reference statement above, what would happen to invoices in the Invoice table that are linked to a customer **if that customer is deleted**.

- a. Those invoices would remain in the database.
- b. Those invoices would be deleted also.
- c. The Customer ID for those invoices would be changed to NULL.
- d. Nothing would happen.

<u>Customer ID (PK)</u> Customer Last Name Customer First Name Street Address City State Zip
--

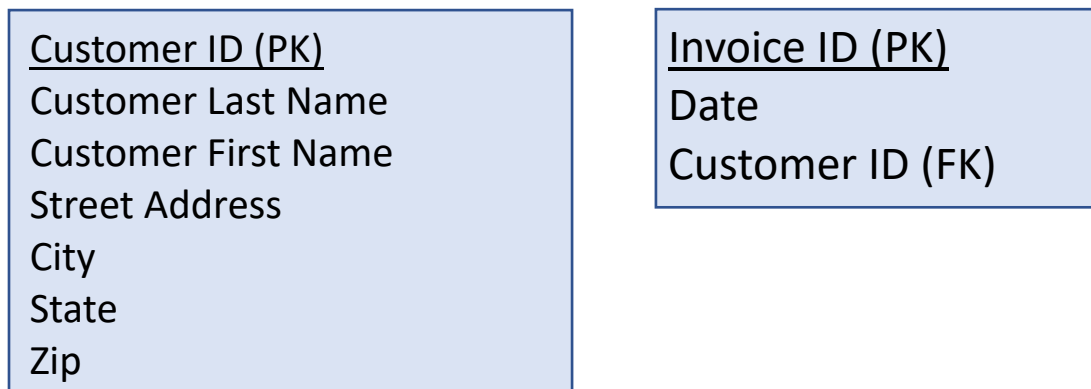
<u>Invoice ID (PK)</u> Date Customer ID (FK)
--

8.

```
CREATE TABLE Invoice (  
    invoice_id      INT      NOT NULL  AUTO_INCREMENT,  
    date            DATE     NOT NULL,  
    customer_id     INT      NOT NULL,  
    PRIMARY KEY (invoice_id),  
    FOREIGN KEY (customer_id) REFERENCES Customer (customer_id) ON DELETE  
RESTRICT  
);
```

Looking at the Customer and Invoice tables and the CREATE TABLE for the Invoice table with foreign key reference statement above, what would happen to invoices in the Invoice table that are linked to a customer **if that customer is deleted**.

- a) Those customers would remain in the customer database.
- b) Those invoices would be deleted also.
- c) The Customer ID for those invoices would be changed to NULL.
- d) The delete of the invoices would not be allowed.



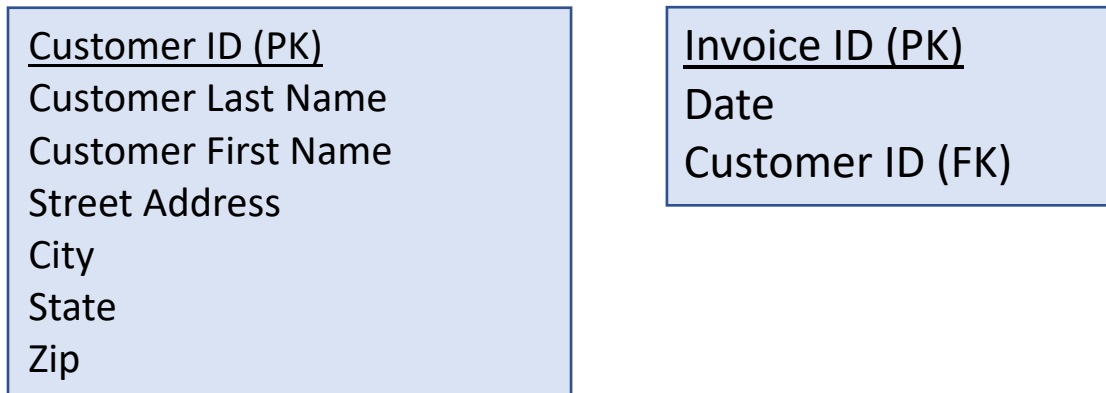
9.

```
CREATE TABLE Invoice (  
    invoice_id      INT      NOT NULL  AUTO_INCREMENT,  
    date            DATE     NOT NULL,  
    customer_id     INT      NOT NULL,
```

PRIMARY KEY (invoice_id),
FOREIGN KEY (customer_id) REFERENCES Customer (customer_id) **ON DELETE SET TO NULL**
);

Looking at the Customer and Invoice tables and the CREATE TABLE for the Invoice table with foreign key reference statement above, what would happen to invoices in the Invoice table that are linked to a customer if that customer is deleted.

- a. Those invoices would remain in the database.
- b. Those invoices would be deleted also.
- c. The Customer ID for those invoices would be changed to NULL.
- d. The delete of the Customer would not be allowed.



10.

Which of the following are true about materialized view (Choose 2)?

- a. It is a base table.
- b. It is stored.
- c. It must be refreshed whenever the base table changes.
- d. The results are stored as a temporary table.

11.

The Customer table will have the following columns:

CustomerID—positive integer

FirstName—variable-length string with up to 50 characters

MiddleInitial—fixed-length string with 1 character

LastName—variable-length string with up to 50 characters

DateOfBirth—date

CreditLimit—positive decimal value representing a cost of up to \$19,999.99 with 2 digits for cents

Write a SQL statement to create the Customer table.
Do not add any additional constraints to any column beyond what is stated.

12.

The Genre table has the following columns:

GenreCode—variable-length string, primary key

GenreDescription—variable-length string

The Book table should have the following columns:

Title—variable-length string, maximum 30 characters

GenreCode—variable-length string, maximum 5 characters

Write a SQL statement to create the Book table. Designate the GenreCode column in the Book table as a foreign key to the GenreCode column in the Genre table.

13.

The Automobile table has the following columns:

ID—integer, primary key

Make—variable-length string

Model—variable-length string

Year—integer

A new column must be inserted into the Automobile table:

Column name: SafetyRating

Data type: A positive number that is less than 80 with 1 decimal space.

Write a SQL statement to add the SafetyRating column to the Automobile table

14.

The Book table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create a view named MyBooks that contains the Title, Genre, and Year columns for all books. Ensure your result set returns the columns in the order indicated.

15.

A database has a view named BookView.

Write a SQL statement to delete the view named BookView from the database.

16.

The Book table has the following columns:

ID—integer

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to modify the Book table to make the ID column the primary key

17.

The Book table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

The YearSales table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer

Write a SQL statement to designate the Year column in the Book table as a foreign key to the Year column in the YearSales table.

18.

The Book table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create an index named idx_year on the Year column of the Book table.

19.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

The following data needs to be added to the Book table:

Title	Genre	Year
The Joy Luck Club	No Data	1989

Write a SQL statement to insert the indicated data into the Book table.

20.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to delete the row with the ID value of 3 from the Book table.

21.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to update the Year value to be 2022 for all books with a Year value of 2020.

22. Which query illustrates performing an outer join of the Book table with a different table?

- SELECT B.Title, A.Author FROM Book B, Author A
WHERE B.AuthorID = A.AuthorID;
- SELECT B.Title, A.Author FROM Book B
LEFT JOIN Book MB ON B.ID = MB.ID, Author A;
- SELECT Book.Title, A.Author FROM Book B
RIGHT JOIN Author A ON B.AuthorID = A.ID;
- SELECT B.Title, A.Author FROM Book B
INNER JOIN Author A ON B.AuthorID = A.ID;

23. Assume there are two tables, A and B.

Which rows will always be included in the result set if Table A is inner joined with Table B?

- Only rows in Tables A and B that share the join condition
- All rows in Table B
- All rows in Table A
- Only rows in Tables A and B that do not share the join condition.

24.

The database contains a table named Book.

Write a SQL query to return all data from the Book table without directly referencing any column names.

25.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to retrieve the Title and Genre values for all records in the Book table with a Year value of 2020. Ensure your result set returns the columns in the order indicated.

26.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to display all Title values in alphabetical order A–Z

27.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to output the unique Genre values and the number of books with each genre value from the Book table as GenreCount. Sort the results by the Genre in reverse alphabetical order Z-A. Ensure your result set returns the columns in the order indicated.

28 A)

The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

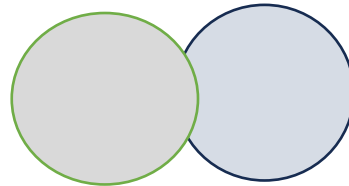
The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer

Write a SQL query to display both the TotalSales and the Title(if available) for all YearSales. Ensure your result set returns the columns in the order indicated.



28 B)

The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

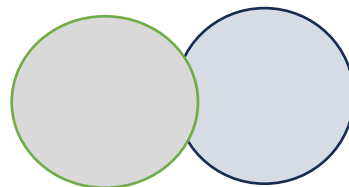
The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer

Write a SQL query to display both the Title and the TotalSales for all books. Ensure your result set returns the columns in the order indicated.



28 C)

The **YearSales** table has the following columns:

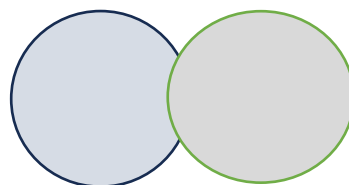
Year—integer

TotalSales—bigint unsigned

Releases—integer

The **Book** table has the following columns:

ID—integer, primary key, auto_increment



Title—variable-length string
Genre—variable-length string
Year—integer

Write a SQL query to display both the Title and TotalSales regardless of whether they had any sales for all Books. Ensure your result set returns the columns in the order indicated.

29.

The Book table has the following columns:
ID—integer, primary key, auto_increment
Title—variable-length string
Genre—variable-length string
Year—integer
Cost – positive decimal value up to \$9999.99

Write a SQL statement that will **add \$1 (for charity) to the cost of any book** that costs \$50 dollars or less.

30.

The Book table has the following columns:
ID—integer, primary key, auto_increment
Title—variable-length string
Genre—variable-length string
Year—integer
Write a SQL query to return how many books have a Year value of 2019.

31.

The **Movie** table has the following columns:
MovieID—integer, primary key
Title—variable-length string
ReleaseDate – date
Cost – integer, positive
StreamID— fixed length character

A) Write a SQL statement to pull how much the most expensive movie cost to make.

B) Write another SQL statement to pull all data for movies that were above average cost to make.

32.

The Employee table has the following columns:
EmployeeID – integer, primary key
FirstName – variable-length character string
LastName – variable-length character string

EmployeeType – variable-length character string
Salary – decimal, positive

Write a SQL statement to pull the **EmployeeType** and the **highest salary by EmployeeType** in the Employee table.

D427 Practice Test 1 ANSWER KEY

Grade this test all or nothing like the OA does! No partial credit is given on the OA (even in SQL coding questions.) You must score 17/25 on the OA to pass, so try to score 90% on these practice tests & PreA, to be confident you will pass!

1. Which of the following functions would be used to combine two columns:

- a. TRUNCATE()
- b. AGGREGATE()
- c. SUBSTRING()
- d. CONCAT()

2. The Book table has the following columns:

genre – varchar(20)

pages – integer

author_id – char(3)

isbn_number – varchar(20)

Which column should be designated as the primary key for the Book table?

- a. genre
- b. pages
- c. author_id
- d. isbn_number

3. The Book table has the following columns:

genre – varchar(20)

pages – integer

author_id – char(3)

isbn_number – varchar(20)

Which column should be designated as the foreign key for the Book table?

- a. genre
 - b. pages
 - c. author_id
 - d. isbn_number
4. Which data type represents numbers with fractional values:

- a. Integer
- b. Decimal
- c. Character
- d. Binary

5. Which of the following is a DDL commands?

- a. INSERT
- b. SELECT
- c. CREATE INDEX
- d. UPDATE

6. Which of the following is a DML command?

- a. CREATE VIEW
- b. CREATE TABLE
- c. INSERT
- d. ALTER INDEX

7.

```
CREATE TABLE Invoice (  
    invoice_id      INT      NOT NULL  AUTO_INCREMENT,  
    date            DATE     NOT NULL,  
    customer_id     INT      NOT NULL,  
    PRIMARY KEY (invoice_id),  
    FOREIGN KEY (customer_id) REFERENCES Customer (customer_id) ON DELETE CASCADE  
);
```

Looking at the Customer and Invoice tables and the CREATE TABLE for the Invoice table with foreign key reference statement above, what would happen to invoices in the Invoice table that are linked to a customer if that customer is deleted.

- a. Those invoices would remain in the database.
- b. Those invoices would be deleted also.
- c. The Customer ID for those invoices would be changed to NULL.

d. Nothing would happen.

Customer ID (PK)
Customer Last Name
Customer First Name
Street Address
City
State
Zip

Invoice ID (PK)
Date
Customer ID (FK)

8.

```
CREATE TABLE Invoice (  
  invoice_id      INT      NOT NULL  AUTO_INCREMENT,  
  date            DATE     NOT NULL,  
  customer_id     INT      NOT NULL,  
  PRIMARY KEY (invoice_id),  
  FOREIGN KEY (customer_id) REFERENCES Customer (customer_id) ON DELETE  
RESTRICT  
);
```

Looking at the Customer and Invoice tables and the CREATE TABLE for the Invoice table with foreign key reference statement above, what would happen to invoices in the Invoice table that are linked to a customer if that customer is deleted.

- a) Those customers would remain in the customer database.
- b) Those invoices would be deleted also.
- c) The Customer ID for those invoices would be changed to NULL.
- d) The delete of the invoices would not be allowed.

Customer ID (PK)
Customer Last Name
Customer First Name
Street Address
City
State
Zip

Invoice ID (PK)
Date
Customer ID (FK)

9.

```
CREATE TABLE Invoice (
```

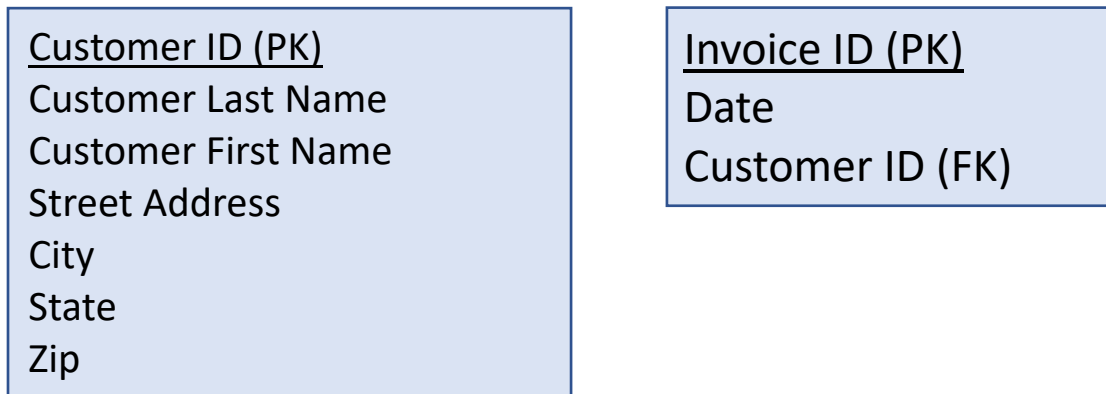
```

invoice_id      INT      NOT NULL  AUTO_INCREMENT,
date            DATE     NOT NULL,
customer_id     INT      NOT NULL,
PRIMARY KEY (invoice_id),
FOREIGN KEY (customer_id) REFERENCES Customer (customer_id) ON DELETE SET TO
NULL
);

```

Looking at the Customer and Invoice tables and the CREATE TABLE for the Invoice table with foreign key reference statement above, what would happen to invoices in the Invoice table that are linked to a customer if that customer is deleted.

- a. Those invoices would remain in the database.
- b. Those invoices would be deleted also.
- c. The Customer ID for those invoices would be changed to NULL.
- d. The delete of the Customer would not be allowed.



10.

Which of the following are true about materialized view (Choose 2)?

- a. It is a base table.
- b. It is stored.
- c. It must be refreshed whenever the base table changes.
- d. The results are stored as a temporary table.

11.

The Customer table will have the following columns:

CustomerID—positive integer

FirstName—variable-length string with up to 50 characters

MiddleInitial—fixed-length string with 1 character

LastName—variable-length string with up to 50 characters

DateOfBirth—date

CreditLimit—positive decimal value representing a cost of up to \$19,999, with 2 digits for cents

Write a SQL statement to create the Customer table.

Do not add any additional constraints to any column beyond what is stated.

```
CREATE TABLE Customer (  
    CustomerID INT UNSIGNED,  
    FirstName VARCHAR(50),  
    MiddleInitial CHAR(1),  
    LastName VARCHAR(50),  
    DateOfBirth DATE,  
    CreditLimit DECIMAL(7,2) UNSIGNED CHECK(CreditLimit <= 19999.99)  
);
```

12.

The Genre table has the following columns:

GenreCode—variable-length string, primary key

GenreDescription—variable-length string

The Book table should have the following columns:

Title—variable-length string, maximum 30 characters

GenreCode—variable-length string, maximum 5 characters

Write a SQL statement to create the Book table. Designate the GenreCode column in the Book table as a foreign key to the GenreCode column in the Genre table.

```
CREATE TABLE Book (  
    Title VARCHAR(30),  
    GenreCode VARCHAR(5),  
    FOREIGN KEY (GenreCode) REFERENCES Genre(GenreCode)  
);
```

13.

The Automobile table has the following columns:

ID—integer, primary key

Make—variable-length string

Model—variable-length string

Year—integer

A new column must be added to the Automobile table:

Column name: SafetyRating

Data type: A positive number that is less than 80 with 1 decimal space.

Write a SQL statement to add the SafetyRating column to the Automobile table.

ALTER TABLE Automobile

ADD SafetyRating DECIMAL(3,1) UNSIGNED CHECK(SafetyRating < 80.0);

OR

ALTER TABLE Automobile

ADD SafetyRating DECIMAL(3,1) UNSIGNED CHECK(SafetyRating <= 79.9);

14.

The Book table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create a view named MyBooks that contains the Title, Genre, and Year columns for all books. Ensure your result set returns the columns in the order indicated.

CREATE VIEW MyBooks AS

SELECT Title, Genre, Year

FROM Book;

15.

A database has a view named BookView.

Write a SQL statement to delete the view named BookView from the database.

DROP VIEW BookView;

16.

The Book table has the following columns:

ID—integer,

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to **modify** the **Book table** to make the **ID column** the primary key.

ALTER TABLE Book
ADD PRIMARY KEY (ID);

17.

The Book table has the following columns:

ID—integer, primary key
Title—variable-length string
Genre—variable-length string
Year—integer

The YearSales table has the following columns:

Year—integer
TotalSales—bigint unsigned
Releases—integer

Write a SQL statement to designate the Year column in the Book table as a foreign key to the Year column in the ~~TotalSales~~ ^{YearSales table.} table.

ALTER TABLE Book
ADD FOREIGN KEY (Year) REFERENCES YearSales (Year);

18.

The Book table has the following columns:

ID—integer, primary key
Title—variable-length string
Genre—variable-length string
Year—integer

Write a SQL statement to create an index named idx_year on the Year column of the Book table.

CREATE INDEX idx_year
ON Book(Year);

19.

The Book table has the following columns:

ID—integer, primary key, auto_increment
Title—variable-length string
Genre—variable-length string
Year—integer

The following data needs to be added to the Book table:

Title	Genre	Year
The Joy Luck Club	No Data	1989

Write a SQL statement to insert the indicated data into the Book table.

**INSERT INTO Book (Title, Genre, Year) VALUES
(‘The Joy Luck Club’, NULL, 1989);**

20.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to delete the row with the ID value of 3 from the Book table.

**DELETE FROM Book
WHERE ID = 3;**

21.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to update the Year value to be 2022 for all books with a Year value of 2020.

**UPDATE Book
SET Year = 2022
WHERE Year = 2020;**

22. Which query illustrates performing an outer join of the Book table with a different table?

- a. SELECT B.Title, A.Author FROM Book B, Author A
WHERE B.AuthorID = A.AuthorID;
- b. SELECT B.Title, A.Author FROM Book B
LEFT JOIN Book MB ON B.ID = MB.ID, Author A
- c. SELECT Book.Title, A.Author FROM Book B
RIGHT JOIN Author A ON B.AuthorID = A.ID

d. `SELECT B.Title, A.Author FROM Book B
INNER JOIN Author A ON B.AuthorID = A.ID`

This question is asking which statement illustrates an OUTER JOIN, not an INNER JOIN. The only 2 statements above that include outer joins are b and c (LEFT JOIN and RIGHT JOIN).

B is an invalid statement, so C has to be the answer. Please review the JOINS PPT to help you!

23.

Assume there are two tables, A and B.

Which rows will always be included in the result set if Table A is inner joined with Table B?

- a. Only rows in Tables A and B that share the join condition
- b. All rows in Table B
- c. All rows in Table A
- d. Only rows in Tables A and B that do not share the join condition.

24.

The database contains a table named Book.

Write a SQL query to return all data from the Book table without directly referencing any column names.

SELECT * FROM Book;

25.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to retrieve the Title and Genre values for all records in the Book table with a Year value of 2020. Ensure your result set returns the columns in the order indicated.

SELECT Title, Genre

FROM Book

WHERE Year = 2020;

26.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to display all Titles in alphabetical order from A–Z.

This question is testing to see if the student knows how to use the ORDER BY clause. Here is the general syntax:

```
SELECT column1, column2, ...  
FROM table_name  
ORDER BY column1, column2, ... ASC|DESC;
```

And here is the correct answer.

```
SELECT Title  
FROM Book  
ORDER BY Title ASC;
```

27.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to output the unique Genre values and the number of books with each genre value from the Book table as GenreCount. Sort the results by the Genre in reverse alphabetical order Z-A. Ensure your result set returns the columns in the order indicated.

In this case you are being asked to count the number of Movies having each type of Genre. For this reason, you must use GROUP BY to group all moves with each Genre together and then do a count. The attribute you are grouping on must be listed first.

```
SELECT DISTINCT Genre, COUNT(*) AS GenreCount  
FROM Book  
GROUP BY Genre  
ORDER BY Genre DESC;
```

Please take a look at these resources:

[GROUP BY](#)
[COUNT](#)

28 A)

The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

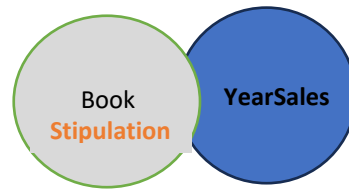
Year—integer

The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer



Write a SQL query to display both the TotalSales and the Title (**if available**) for all YearSales. Ensure your result set returns the columns in the order indicated.

```
SELECT YearSales.TotalSales, Book.Title
FROM Book
RIGHT JOIN YearSales
ON Book.Year = YearSales.Year;
```

Here, you must have a **RIGHT JOIN** because the stipulation “**if available**” comes from the left table (stipulation=some so bring SOME of these), and TotalSales for **ALL YearSales (RIGHT Table.)**

Review the JOINS and Aggregate Functions PPT in Slide Show Mode.

28 B)

The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

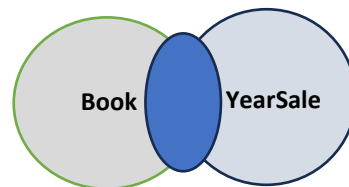
Year—integer

The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer



Write a SQL query to display both the Title and the TotalSales for all books. Ensure your result set returns the columns in the order indicated.

```
SELECT Book.Title, YearSales.TotalSales
FROM Book
INNER JOIN YearSales
ON Book.Year = YearSales.Year;
```

This one is an **INNER JOIN** because there are **no stipulations** (if available/regardless of/include nulls). It really shouldn't say for ALL books, but it's what Zybooks does! It should say for ANY Books or ANY YearSales. We are trying to get this fixed with the publisher of Zybooks. Please review the **JOINS and Aggregate Functions PPT in Slide Show Mode** for more in-depth explanations and practice.

28 C)

The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer

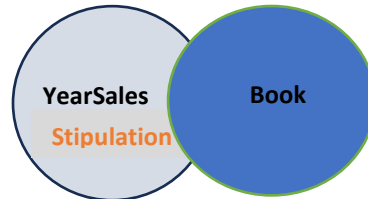
The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer



Write a SQL query to display both the Title and TotalSales **regardless of whether they had any sales** for all Books. Ensure your result set returns the columns in the order indicated.

```
SELECT Book.Title, YearSales.TotalSales
FROM YearSales
RIGHT JOIN Book
ON YearSales.Year = Book.Year;
```

Here, you must AGAIN have a **RIGHT JOIN** because of the **stipulation “regardless of whether they had any sales”** comes from the LEFT table (stipulation=some -> bring some of these from the left table), and **Title for ALL BOOKS**. Review the JOINS and Aggregate Functions PPT in Slide Show Mode.

29.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Cost – positive decimal value up to \$9999.99.

Write a SQL query that will **add \$1 (for charity) to the cost of any book** that costs \$50 dollars or less.

```
UPDATE Book
```

SET Cost = Cost + 1
WHERE Cost <= 50;

30.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to return how many books have a Year value of 2019.

SELECT COUNT(*)
FROM Book
WHERE Year = 2019;

31.

The **Movie** table has the following columns:

MovieID—integer, primary key

Title—variable-length string

ReleaseDate – date

Cost – integer, positive

StreamID— fixed length character

A) Write a SQL statement to pull how much the most expensive movie cost to make.

SELECT MAX(Cost)
FROM Movie;

B) Write another SQL statement to pull all data for movies that were **above average cost** to make.

SELECT *
FROM Movie
WHERE Cost >
(SELECT AVG(Cost) FROM Movie);

This is a subquery and must be used because you cannot put an Aggregate Function in a WHERE clause directly. You must use a subquery so look for ‘below average’ or ‘above average’ in the question.

32.

The Employee table has the following columns:

EmployeeID – integer, primary key

FirstName – variable-length character string

LastName – variable-length character string

EmployeeType – variable-length character string

Salary – decimal, positive

Write a SQL statement to pull the **EmployeeType** and the **highest salary by EmployeeType** in the Employee table:

```
SELECT EmployeeType, MAX(Salary)
FROM Employee
GROUP BY EmployeeType;
```

Here is what the result could look like:

Employee Type	Max(Salary)
Associate	60000
Supervisor	80000
Manager	100000
Director	120000